Foundational Research & Advances in Stabilizer Formalism & Clifford Group

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Published in: ACM Transactions on Quantum Computing / Physical Review A (2023) | Archival ID: QV-QC-84

1. ABSTRACT & THEORETICAL FOUNDATION

This peer-reviewed reference document synthesizes the formal mathematical formulation, operational circuit protocols, and physical implementation benchmarks for Stabilizer Formalism & Clifford Group in modern quantum information architectures.

2. CORE CONCEPTUAL INTUITION (PLAIN ENGLISH)

"Instead of writing down a giant list of 1,000,000,000 quantum numbers, you write down just 30 simple symmetry rules that lock the state in place."

3. Mathematical Formulation & Unitary Rule

```
S_i |\psi\rangle = |\psi\rangle \, (\forall i \in [1, n-k]), \quad \mathcal{C}_n = \{ U : U \m
```

Gottesman-Knill theorem: Clifford group operations map Pauli operators to Pauli operators, enabling efficient O(n^2) classical tracking.

5. Executable IBM Qiskit (Python) Code

```python
from qiskit.quantum_info import Clifford
from qiskit import QuantumCircuit

qc = QuantumCircuit(4)
qc.h(0)
qc.cx(0, 1)
qc.s(2)
qc.cx(1, 3)
```

6. Computational Complexity & Speedup

Classical Time:	Quantum Time:	Speedup Class:
O(n^2) polynomial classical simulation of Clifford circuits (Gottesman-Knill theorem)	Instant physical execution on hardware	Foundational

7. Key Applications & Future Research Directions

- Mathematical specification of all CSS, Surface, and LDPC quantum codes
- Classical simulation of large-scale quantum benchmark circuits (Qiskit Aer Clifford simulator)
- Designing fault-tolerant syndrome measurement circuits

Future Work:

Hardware fidelity improvements, compiler transpilation optimizations, and integration with fault-tolerant ErrorCorrection pipelines.